

## **TAE -2**

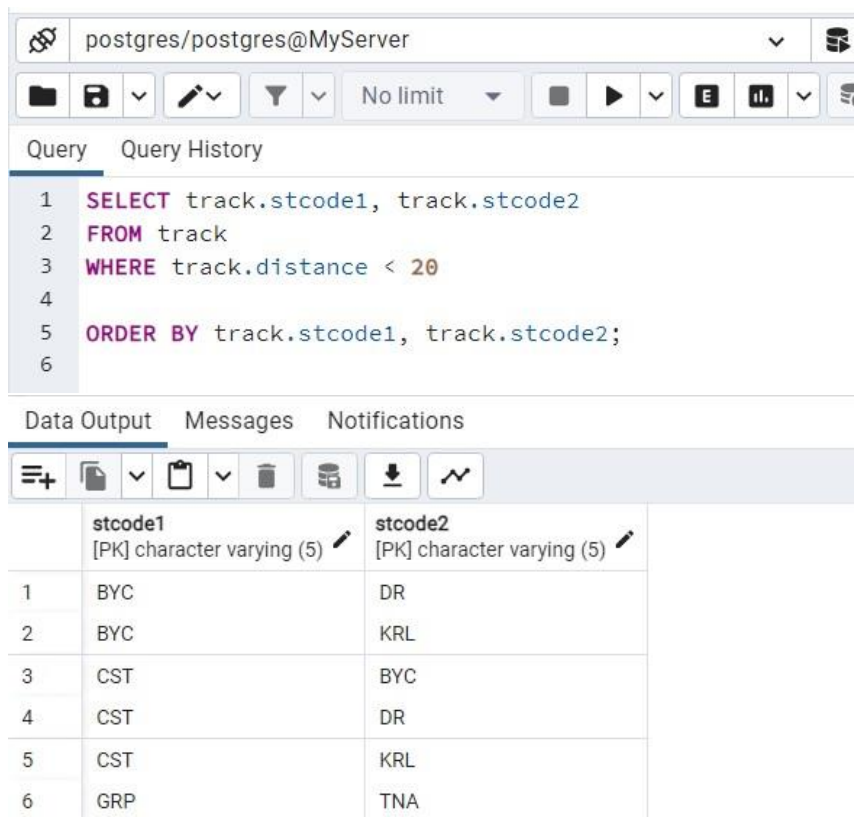
### **Subject: Database Management System (DBMS)**

|         |                |
|---------|----------------|
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| Subject | DBMS Tae 2     |

## Query 1 -

**Find pairs of stations (station codes) that have a track (direct connection) with distance less than 20 Kms between them.**

```
SELECT track.stcode1, track.stcode2
FROM track
WHERE track.distance < 20
ORDER BY track.stcode1, track.stcode2;
```



The screenshot shows a PostgreSQL query editor interface. At the top, the connection is 'postgres/postgres@MyServer'. Below the toolbar, the 'Query' tab is active, displaying the following SQL query:

```
1 SELECT track.stcode1, track.stcode2
2 FROM track
3 WHERE track.distance < 20
4
5 ORDER BY track.stcode1, track.stcode2;
6
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query in a table format. The table has two columns: 'stcode1' and 'stcode2', both of type '[PK] character varying (5)'. The results are as follows:

|   | stcode1<br>[PK] character varying (5) | stcode2<br>[PK] character varying (5) |
|---|---------------------------------------|---------------------------------------|
| 1 | BYC                                   | DR                                    |
| 2 | BYC                                   | KRL                                   |
| 3 | CST                                   | BYC                                   |
| 4 | CST                                   | DR                                    |
| 5 | CST                                   | KRL                                   |
| 6 | GRP                                   | TNA                                   |

Query

2 -

**Find the IDs of all the trains which have a stop at THANE.**

```
SELECT t.id
FROM trainhalts AS t
JOIN station AS s ON t.stcode = s.stcode
WHERE s.name = 'THANE'
```

| Query |                                          | Query History |  |
|-------|------------------------------------------|---------------|--|
| 1     | SELECT t.id                              |               |  |
| 2     | FROM trainhalts AS t                     |               |  |
| 3     | JOIN station AS s ON t.stcode = s.stcode |               |  |
| 4     | WHERE s.name = 'THANE'                   |               |  |

| Data Output |                       | Messages | Notifications |
|-------------|-----------------------|----------|---------------|
|             |                       |          |               |
|             | id                    |          |               |
|             | character varying (5) |          |               |
| 1           | KP11                  |          |               |
| 2           | A65                   |          |               |

### Query 3 -

**Find the names of all trains that start at MUMBAI.**

```
SELECT DISTINCT t.name
FROM train AS t
WHERE t.id IN (
    SELECT th.id
    FROM trainhalts AS th
    JOIN station AS s ON th.stcode = s.stcode
    WHERE s.name = 'MUMBAI' AND th.seqno = 0
);
```

Query

Query History

1

2

3

4

5

6

7

8

SELECT DISTINCT t.name

FROM train AS t

WHERE t.id IN (

SELECT th.id

FROM trainhalts AS th

JOIN station AS s ON th.stcode = s.stcode

WHERE s.name = 'MUMBAI' AND th.seqno = 0

);

Data Output

Messages

Notifications

≡+

▼

▼

name

character varying (20)

1

CST-AMR\_LOCAL

2

CST-KYN

#### Query 4 -

**List all the stations in order of visit by the train 'CST-AMR\_LOCAL'.**

```
SELECT s.name
FROM station AS s
JOIN trainhalts AS th ON th.stcode = s.stcode
WHERE th.id = 'A65'
ORDER BY th.seqno;
```

| Query |                                                                    | Query History |
|-------|--------------------------------------------------------------------|---------------|
| 1     | <b>SELECT</b> s.name                                               |               |
| 2     | <b>FROM</b> station <b>AS</b> s                                    |               |
| 3     | <b>JOIN</b> trainhalts <b>AS</b> th <b>ON</b> th.stcode = s.stcode |               |
| 4     | <b>WHERE</b> th.id = 'A65'                                         |               |
| 5     | <b>ORDER BY</b> th.seqno;                                          |               |

| Data Output  |                        | Messages | Notifications |
|--------------|------------------------|----------|---------------|
| <div> </div> |                        |          |               |
|              | <b>name</b>            |          |               |
|              | character varying (20) |          |               |
| 1            | MUMBAI                 |          |               |
| 2            | BYCULLA                |          |               |
| 3            | DADAR                  |          |               |
| 4            | KURLA                  |          |               |
| 5            | GHATKOPAR              |          |               |
| 6            | THANE                  |          |               |
| 7            | DOMBIVALI              |          |               |
| 8            | KALYAN                 |          |               |
| 9            | AMBARNATH              |          |               |

## Query 5 -

**Find the name of the trains which have stop at Thane, before the 6th station in the route of the train.**

```

SELECT DISTINCT t.name
FROM train AS t
JOIN trainhalts AS th ON t.id = th.id
JOIN station AS s ON th.stcode = s.stcode
WHERE s.stcode = 'TNA' AND th.seqno <= 6

```

Query Query History

```
1 SELECT DISTINCT t.name
2 FROM train AS t
3 JOIN trainhalts AS th ON t.id = th.id
4 JOIN station AS s ON th.stcode = s.stcode
5 WHERE s.stcode = 'TNA' AND th.seqno <= 6
```

Data Output Messages Notifications



|   | name<br>character varying (20) |
|---|--------------------------------|
| 1 | CST-AMR_LOCAL                  |